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Supplementary information

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Using deep learning to generate key variables in global mitigation scenarios

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Supplementary Text S1: Background and overview of climate mitigation scenarios

S1.1 Integrated assessment models and climate mitigation scenarios

Integrated assessment models (IAMs) are simplified representations of complex physical and social systems that focus on the interaction between economy, society and the environment[1]. These models represent the coupled energy-economy-land-climate system to varying degrees. IAMs can endogenously describe changes in emissions resulting from alterations in energy and land use activities, particularly in response to climate action[2]. A key purpose of IAMs is to provide orientation knowledge for the deliberation of future climate action strategies by policymakers, civil society and the private sector. This is achieved by presenting various scenarios towards a range of long-term climate outcomes under broad assumptions about future socio-economic, institutional, and technological developments. Future climate change is a prime example of the application of scenarios. Climate mitigation scenarios are constructed to explore possible climate change futures, including socioeconomic developments, energy and land use, greenhouse gas emissions, changes in temperature and precipitation.

S1.2 Intergovernmental Panel on Climate Change (IPCC) AR6 scenario database

The number of scenarios available in the literature has been steadily growing, from several hundred in 2000 to several thousand by the time of AR6[3]. Given the large number of scenarios and data from a large variety of sources, maintaining a database for the assessment is essential to ensure full documentation and transparency of the assessment. AR6 scenario database (https://zenodo.org/records/7197970) contains 3,131 quantitative scenarios with data on socio-economic development, greenhouse gas emissions, and sectoral transformations across energy, land use, transportation, and industry. These scenarios derive from 191 unique modelling frameworks (including different versions and country setups) from 95+ model families of which 98 were globally comprehensive, 71 national or multi-regional, and 20 sectoral models[4].

The global scenarios have been classified, to the extent possible, by their warming outcome. The definition of the climate categories and the distribution of scenarios in the AR6 database across these categories is shown in **Table S1**. The first four of these categories correspond to the ones used in the IPCC SR1.5[5] while the latter four have been added as part of the AR6 to capture a broader range of warming outcomes. Of the total scenarios submitted, 1,202 could be assessed in their associated climate response (**Table S2**).

S1.3 Limitations of AR6 scenario database

IPCC authors repeatedly acknowledge that scenarios databases are "ensembles of opportunity" and warn that they should not be used as statistical samples[6,7]. Although a large ensemble of scenarios was gathered, it should be recognized that only a portion of the full uncertainty space is investigated. This introduces "biases" in the ensemble, for example,

1) The topic of scenario studies included in the database influences the coverage of the scenario space, with extensive model-comparison studies placing significant emphasis on selected topics while less explored topics are addressed by individual models.

2) Scenarios of representative models occupy the majority in the database, and the scenarios produced for some specific projects dominate the database[8,9]. As a result, the statistical analysis of the database provides an assessment of only a subset of the scenarios in the published literature.

3) only 'optimistic' models (i.e., models finding lower mitigation costs) reach the lowest mitigation targets[10]. Where appropriate, sampling bias was recognized in the assessment, but formal methods to reduce bias were not employed due to conceptual limitations[4].

4) Furthermore, although attempts have been made to elicit scenario likelihoods from expert knowledge[11], scenarios are difficult to associate with probabilities as they typically describe a situation of deep uncertainty. This and the non-statistical nature of the scenario ensemble collected in the database do not allow a probabilistic interpretation of the distribution of output variables in the scenario database.

Supplementary Text S2: Machine learning models for scenario grouping and feature identification

S2.1 Preprocessing

To aggregate categories C1-C8, we focus on 1,202 global scenarios in the IPCC AR6 database that passed the vetting process, ensuring these scenarios meet reporting standards and contain sufficient emissions data to provide comprehensive climate outcomes across categories C1 to C8. First, we fill in the missing values using the variable mean for the corresponding year. Second, we perform standardization for the dataset. We can view the dataset as a time series with a time step of 9 (for every 10 years from 2020 to 2100). for a time series $a = (a_1, \ldots a_9)$, with mean $\bar{a}$ and standard deviation a_{std}, the standardized time series $\tilde{a} = (\tilde{a}_1, \ldots \tilde{a}_9)$ is calculated as $\tilde{a}_i = \frac{a_i - \bar{a}}{a_{std}}, i = 1,..9$.

S2.2 K-means clustering algorithm

K-Means is an unsupervised machine learning algorithm based on distance metrics, aiming to partition a set of data points into K distinct clusters such that each data point is assigned to the nearest cluster center. The algorithm follows these specific steps: Initially, K data points are randomly selected as the initial cluster centers. Then, for each data point, the distance to each of the K cluster centers is computed (typically using Euclidean distance), and the data point is assigned to the nearest cluster center. This step is referred to as the "assignment" step. Subsequently, the algorithm updates the cluster centers by calculating the mean of all data points within each cluster and setting this mean as the new cluster center. This step is known as the "update" step. The algorithm iterates between the "assignment" and "update" steps until the cluster centers no longer change or a maximum number of iterations is reached. This iterative process ensures that each data point is assigned to the nearest cluster center, thereby minimizing the total distance from each data point to its respective cluster center.

We selected the Kyoto_Gases feature as the basis for calculating distances in the clustering algorithm. It is difficult to distinguish between neighboring classes using only the Kyoto_Gases feature by UMAP [12] method. Therefore, we decided to aggregate the classes, which would improve the separation between the classes. After aggregating the true categories, the distinction between the categories increased. We then utilized the K-means algorithm to partition the data into three clusters based on UMAP low-dimensional embedding, which resulted in better experimental performance, as shown

in **Figure S1** (the left side represents the distribution of the true data after class aggregation, and the right side shows the clustering results of the K-means algorithm).

First, we use the *cdist_dtw* function from the *tslearn* library to calculate the Dynamic Time Wrapping (DTW) distance between any pair of trajectories to quantify the similarities between scenarios. Second, we use Uniform Manifold Approximation and Projection (UMAP), with *n_neighbors*, *metric* and *random_state* set to 15, precomputed and 30, to reduce the dimensionality of these DTW matrices. This process projects each trajectory onto a two-dimensional plane, where closer data points indicate stronger similarities or relationships between the scenarios. Finally, we implement K-means clustering on the UMAP-reduced space, iteratively reducing *n_clusters* from the original 8 IPCC categories to 3, which in general corresponds to a direct aggregation of C1234(C1-C4), C56(C5-C6), and C78(C7-C8). We also performed our analysis using the original eight IPCC categories, and model performance is significantly decreased due to small sample sizes. **Figure S1** displays the distribution of the first two primary components (UMAP1 and UMAP2) under the original 8 IPCC categories. The data points belonging to similar categories tend to cluster together. Therefore, directly applying the K-means method to this data makes it difficult to distinguish between adjacent categories (e.g., C1, C2, C3, C4).

S2.3 Random Forest and XGBoost classifier

Decision trees are a widely used machine learning algorithm for classification and regression tasks, by recursively partitioning the dataset into smaller subsets. By continuously splitting the dataset with criterion on important features, decision trees can capture complex decision boundaries and ultimately form an interpretable model. However, a single decision tree is prone to overfitting, especially when the dataset is small. To overcome the limitations of a single decision tree, the random forest algorithm was developed. Random forest is an ensemble learning method that builds multiple decision trees and combines their results to improve the model's accuracy and stability. It introduces two levels of randomness: sample randomness and feature randomness. When training each decision tree, random forest constructs the training dataset through sampling, and at each node split, it randomly selects a subset of features to find the best split point. This randomness helps to reduce the model's variance and improve its generalization capability. Additionally, the random forest algorithm can provide insights into the importance of individual features.

Extreme Gradient Boosting (XGBoost) is a powerful machine learning technique for regression and classification tasks. XGBoost builds a series of weak learners, typically

decision trees, to gradually improve the model's predictive performance. The core idea of XGBoost is based on the gradient boosting framework, where new trees are added sequentially to enhance the model's predictions, with each new tree trained on the residuals (i.e., the prediction errors) of the current model. The objective function of XGBoost consists of a loss function and a regularization term. The loss function measures the error between the model's predictions and the true values, while the regularization term is used to control the model's complexity and prevent overfitting. Shapley values, a concept from cooperative game theory, are widely used to fairly allocate the contributions of features in machine learning models, particularly in model interpretation and feature importance analysis. In our experiments, we leverage the SHAP (SHapley Additive exPlanations) library to interpret the XGBoost model and validate the reasonableness of feature selection through different feature importance measures.

Drawing on real-world experience, we initially select 13 key variables. we use the *random forest model* from the *sklearn* library to select feature, and in order to explore the impact of the model itself on the final categories, we also include model features (in the form of one-hot encoding) in the key variables. For each feature, such as Carbon Sequestration|CCS, we use its cumulative amount from 2010-2100 (calculated in the same way as mentioned earlier) as the feature value to participate in the training of the random forest model. As for the model feature, we referred to the article[13] and merge the models, taking the first 9 models from **Table S3**. We use the *train_test_split* fuction from the *sklearn* library, with *test_size*=0.2 and *random_state*=3, to randomly divide the data into training and test sets at an 8:2 ratio. We then train random forest models separately on C1-C8 categories (*n_estimators*=100, *max_depth*=12, *random_state*=5，*min_samples_split*=4) and C1234, C56 and C78 Agg-Category (*n_estimators*=100, *max_depth*=8, *random_state*=8, *min_samples_split*=6).

For the C1-C8 categories, the random forest model achieved an accuracy of 99.37% on the training set and 82.50% on the test set. For the C1234, C56 and C78 categories, the random forest model achieved an accuracy of 98.87% on the training set and 95.00% on the test set. **Figure S3** respectively shows the feature importance results for the two classification standards. Regardless of the classification standard, the importance of the model features is significantly lower compared to the importance of the other key variables. This suggests that the model features have a very small impact on the final classification results. Therefore, we believe that the data distributions of the key variables differ across the various models. The differences in the feature variable data distributions are the primary cause of model bias, rather than being attributed to the models themselves.

Additionally, we refine the 2010-2050 period into two sub-periods: 2010-2030 and 2030-2050. The average changes in features during these two sub-periods were used as new features to train random forest models. We split the dataset into training and testing sets (*test_size*=0.2, *random_state*=3) and train random forest models based on the C1-C8 and C1234, C56 and C78 classification standards, respectively. For the C1-C8 classification (*max_depth*=14, *min_samples_split*=6, *n_estimators*=1000, *random_state*=5), the model achieves accuracy of 99.50% on the training set and 85.00% on the testing set. For the C1234, C56 and C78 classification (*n_estimators*=1000, *max_depth*=10, *min_samples_split*=8, *random_state*=8), the model achieved accuracy of 99.75% on the training set and 95.50% on the testing set. **Figure S4** illustrates the feature importance results under two classification standards. To facilitate observation, we categorize the key variables into four groups: 2010-2030, 2030-2050, Cumulative Amount, and Model. We summarize the feature importance for each category by calculating the total importance of the features within each group (**Figure S5**).

We use another method for measuring feature importance: Shapley values. We can observe whether the key variables identified remained consistent across different measurement methods. We standardize the cumulative amounts of the 13 key variables and their corresponding Kyoto_Gases cumulative amounts using the *StandardScaler* function from the *sklearn* library. Then, we use the *XGBRegressor* function from the *xgboost* library (*objective*=reg:squarederror, *max_depth*=6, *reg_alpha*=0.2, *random_state*=5, *n_estimators*=100) to predict the Kyoto_Gases cumulative amounts based on the cumulative amounts of the 13 key variables. The model achieves a mean squared error of 0.0006 on the training set and 0.0127 on the testing set. **Figure S6** illustrates the feature importance based on shapley values. Among the top five ranked features, *PrimaryEnergy_Coal*, *PrimaryEnergy_Oil*, *PrimaryEnergy_Gas* and *CarbonSequestration* appeare, which aligns with the feature importance results from the random forest method. This consistency further validates the rationality of our feature selection.

Similarly, we use the *XGBRegressor* function from the *xgboost* library (*objective*=reg:squarederror, *max_depth*=6, *reg_alpha*=0.2, *random_state*=5, *n_estimators*=100) to predict the Kyoto_Gases cumulative amount, using the 2010-2030 average change, the 2030-2050 average change, and the cumulative amount of 13 key variables. The model achieves a mean error of 0.0004 on the training set and 0.0146 on the testing set. We then use the *shapley* library to explain the regression model, resulting in **Figure S7**.

Supplementary Text S3: Generative deep learning model for producing climate mitigation scenarios

Since the experiments are primarily conducted using the *tsgm* library[14], there are inherent random factors present during model training. Even though a global random seed is set in TensorFlow, it does not completely eliminate randomness. Therefore, we store various parameters of the generative deep learning model in h5 files. Additionally, when generating data, it is necessary to first generate latent variables. We use the *numpy.random.randn* function from the *NumPy* library to generate these latent variables and set the random seed to 3.

S3.1 Variational AutoEncoder (VAE)

Variational Auto-Encoder (VAE)[15] utilizes the core idea of Auto-Encoder. An Auto-Encoder consists of an encoder and a decoder. The encoder learns to map the data into an effective low-dimensional representation, and the decoder then uses this low-dimensional information to reconstruct the original data. In the VAE model, the encoder outputs the posterior distribution of the latent variables for each sample. Samples are then drawn from this distribution to obtain the low-dimensional representation of the data in the latent space. The decoder then uses this latent representation to reconstruct the original data.

Based on the previous results of machine learning models, We select six key variables: *PrimaryEnergy_Coal*, *PrimaryEnergy_Oil*, *PrimaryEnergy_Gas*, *FinalEnergy_Liquid*, *CarbonSequestration* and *Kyoto Gases*. Each variable corresponds to a time series with a step size of 10 (2010-2100). For the VAE (Variational Autoencoder) model, we divide the dataset into three classes based on the C1234, C56 and C78 classification standard, and train a separate VAE model for each class. The data corresponding to the C1234 class contains 680 samples, the data corresponding to the C56 class contains 308 samples, and the data corresponding to the C78 class contains 172 samples. For each class of data, it is represented in the form of a tensor. Taking the C1234 classification as an example, we save the data as a tensor with a shape of (680, 10, 5). This tensor represents 680 samples, 10 steps and 5 key variables. Then, we use the *TSFeatureWiseScaler* function from the *tsgm* library to map the data into the range of (0, 1). The data normalization step is crucial, as it ensures that the key variables are on a similar scale, which can improve the stability and performance of the training process.

For the different categories, we employ different VAE model architectures and optimization strategies (C1234, C56 and C78 correspond to VAE model vae_C1234,

vae_C56 and vae_C78, Extended Data Table 3 in main text). To ensure the synthetic data matched the real data, we generate data for each category, and apply the inverse transform to map the data back to a scale that aligns with the real-world conditions.

S3.2 Conditional Generative Adversarial Network (CGAN)

Conditional Generative Adversarial Network (CGAN)[16] is a generative deep learning model used to generate data under specific conditions. CGAN is an extension of Generative Adversarial Networks (GAN), where the introduction of conditional variables allows for more flexible and practical control over the generated output. In a CGAN, there are two core components: the generator and the discriminator. The generator takes random noise and conditional variables as input, and tries to generate realistic data that aligns with the given conditions. The discriminator, on the other hand, receives both real data and synthetic data by the generator, and uses the conditional variables to determine the authenticity of the data. The generator and the discriminator are trained adversarially. the generator aims to deceive the discriminator, making it unable to distinguish the synthetic data from the real data, while the discriminator continuously improves its ability to accurately differentiate the real data from the synthetic data.

We represent the data as a tensor of shape (1160, 10, 5) and use the *TSFeatureWiseScaler* function to scale the data to the (0,1) range. Both the discriminator and generator employe the Adam optimizer with ExponentialDecay（*initial_learning_rate*=1e-4, *decay_steps*=200, *decay_rate*=0.95）, and the loss function is binary cross-entropy. The model is trained for a total of 500 epochs with batch size of 100.

S3.3 Recurrent Conditional Generative Adversarial Network (RCGAN)

Recurrent Conditional Generative Adversarial Network (RCGAN)[17] is an extension of Generative Adversarial Network (GAN) specifically designed for generating realistic time-series data. RCGAN utilize recurrent neural network (RNN) in both the generator and the discriminator, while also incorporating auxiliary information to enable conditional generation. In RCGAN, the generator employs an RNN to generate the time-series data. At each time step, the generator receives a random seed and can optionally take additional conditional inputs to guide the generation process. The discriminator, on the other hand, also uses an RNN to process the input time-series data and classify it as either real (extracted from the true data distribution) or synthetic (produced by the generator). The recurrent structure and the conditional generation

capabilities of RCGANs make them well-suited for tasks such as time-series forecasting and the synthesis of other sequential data with specific characteristics or dependencies. The integration of RNNs enables RCGAN models to effectively capture the temporal dynamics and dependencies inherent in time-series data, leading to the generation of more realistic and coherent sequences.

The RCGAN model uses the same data processing method as the CGAN model. For both the discriminator and generator, Adam is used in combination with ExponentialDecay (*initial_learning_rate*=1e-3, *decay_steps*=200, *decay_rate*=0.95). The model is trained for a total of 1200 epochs with batch size of 100.

Supplementary Figures

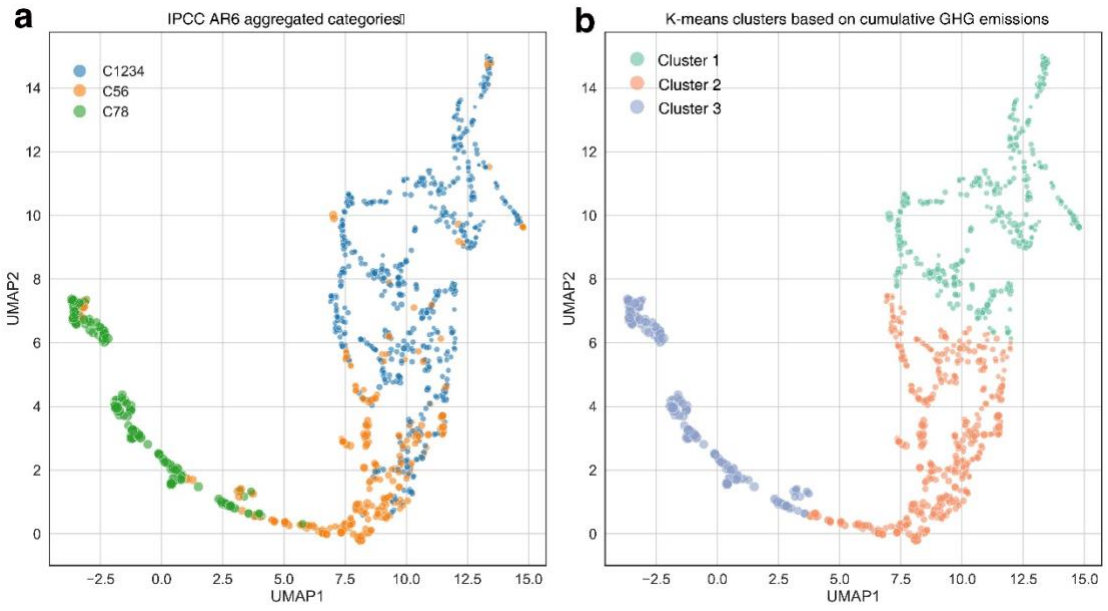


Figure S1. UMAP Projection of different categories

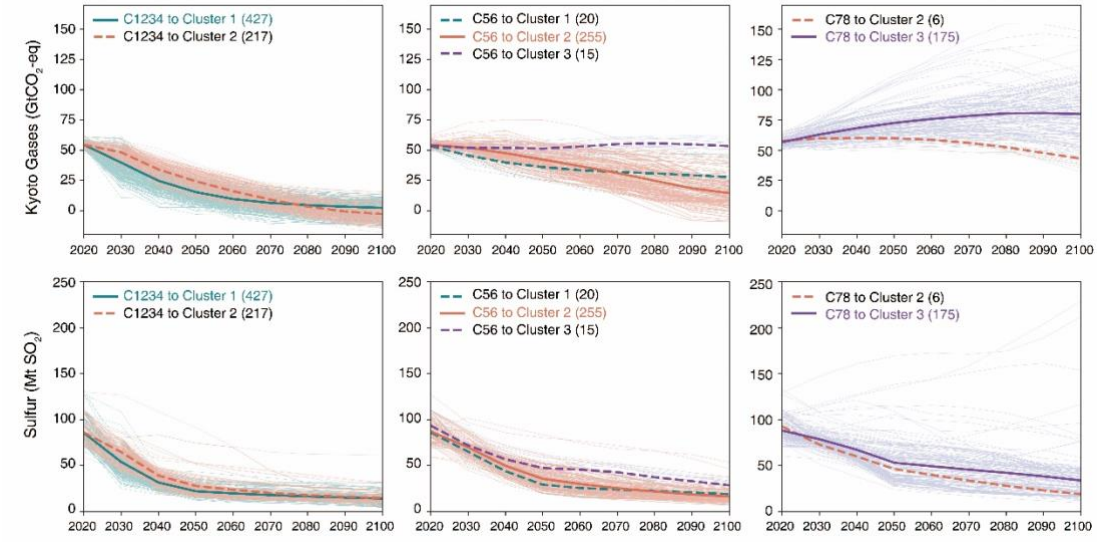


Figure S2. Kyoto gases and SO_2 emissions by aggregated IPCC categories and K-means clusters. Solid lines in each panel are emission trajectories of scenarios mapped to the expected K-means clusters, dashed lines are "mismatched" scenarios mapped to other categories. These discrepancies illustrate significant variability in contributions from climate warming and cooling forcers among scenarios despite similar overall climate outcomes. Some scenarios may have both low Kyoto gas and SO_2 emissions, while others with high levels of both.

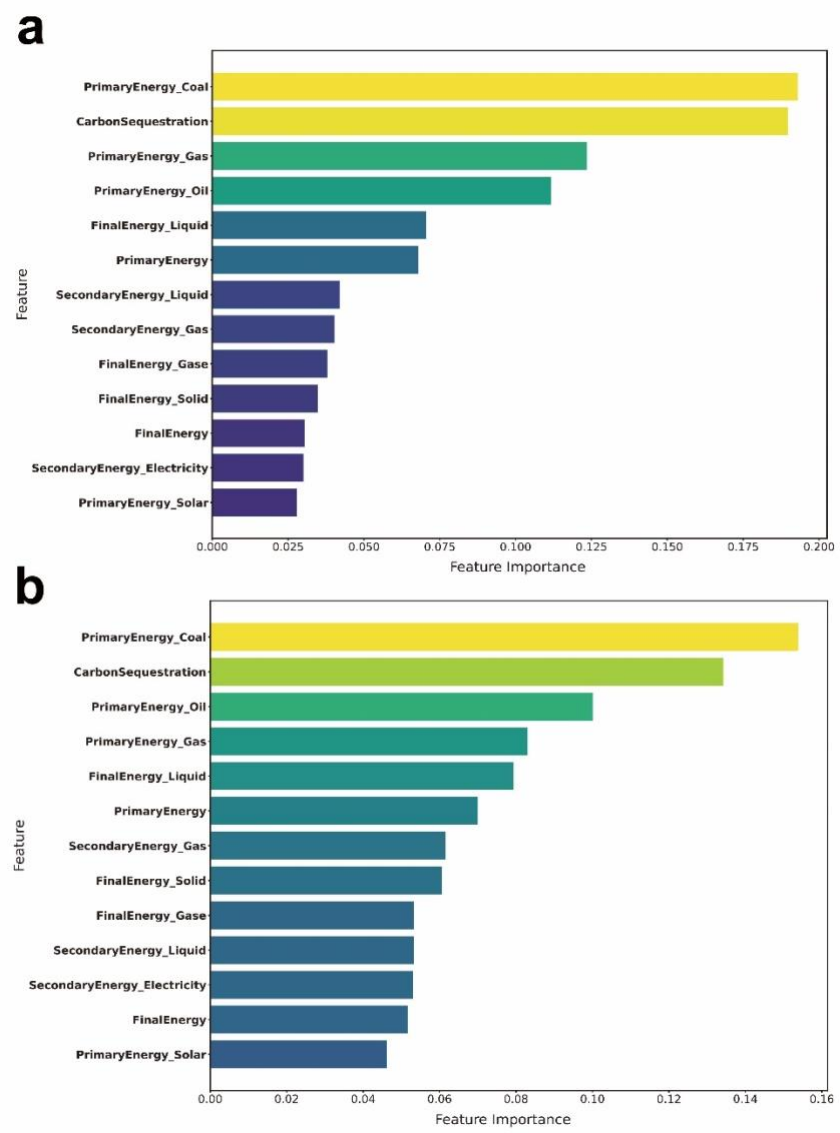


Figure S3. Key variables (without time factor) ranked by feature importance score of random forest model. a. based on aggregated climate mitigation category (C1234, C56, C78). **b.** based on eight IPCC categories (C1-C8)

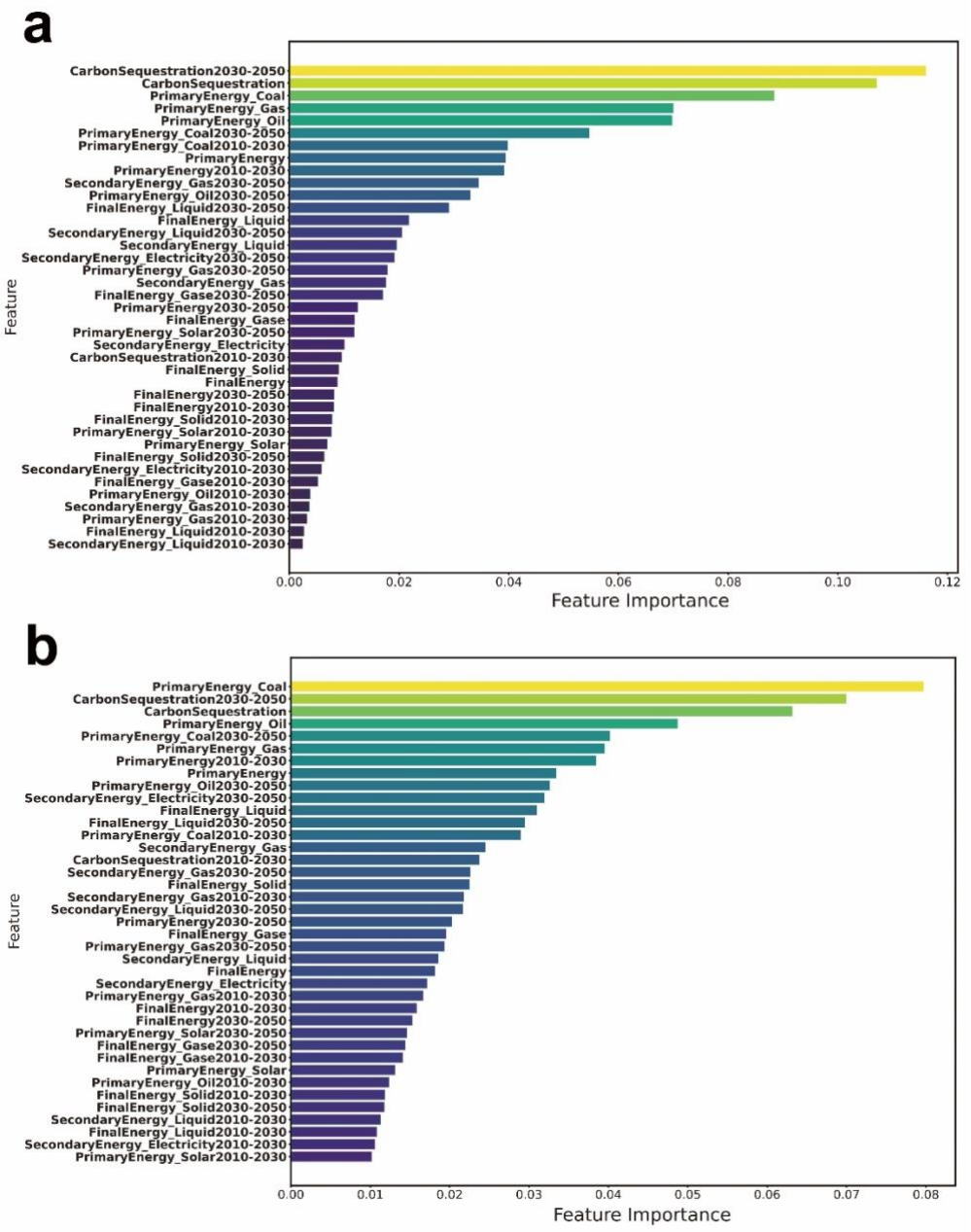


Figure S4. Key variables (with time factor, average change rate during 2010-2030, average change rate during 2030-3050, cumulative amount from 2010 to 2100 and model) ranked by feature importance score of random forest model. a. based on aggregated climate mitigation category (C1234, C56, C78), same as Fig.3 panel a. **b.** based on eight IPCC categories (C1-C8)

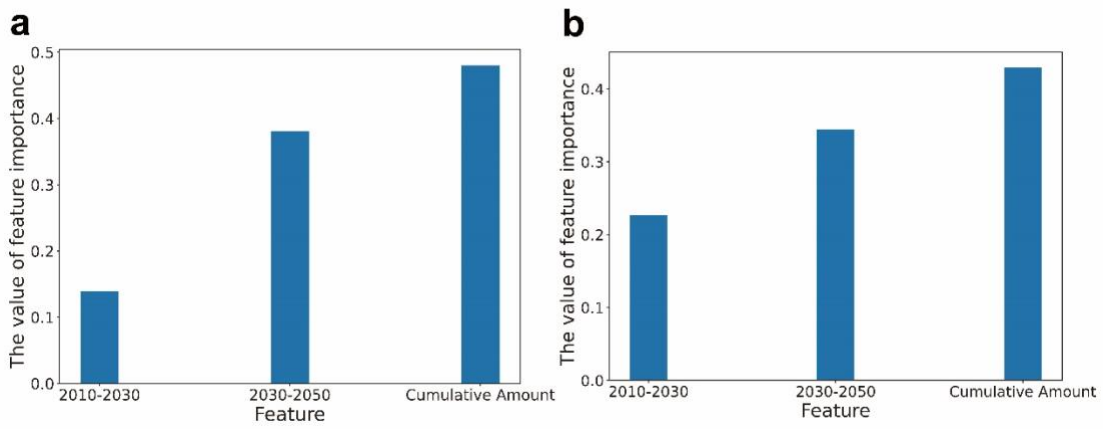


Figure S5. Sum of importance score by variable category (with time factor, average change rate during 2010-2030, average change rate during 2030-3050, cumulative amount from 2010 to 2100 and model). a. based on aggregated climate mitigation category (C1234, C56, C78), same as Fig.2 panel a. **b.** based on eight IPCC categories (C1-C8).

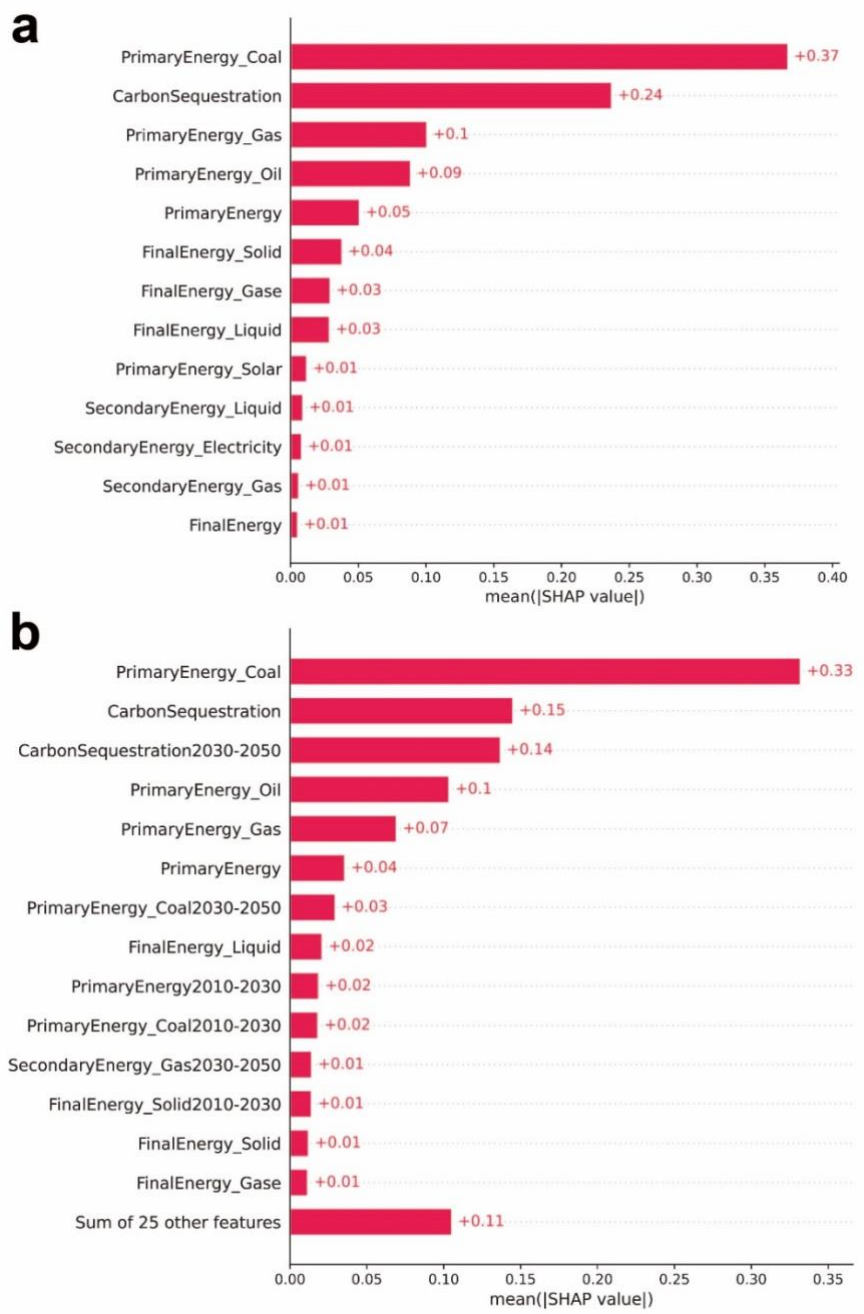


Figure S6. Mean of absolute SHAP value by key variables. a. original 13 key variables. **b.** 39 key variables, including average change rate during 2010-2030, average change rate during 2030-3050, cumulative amount from 2010 to 2100 and models

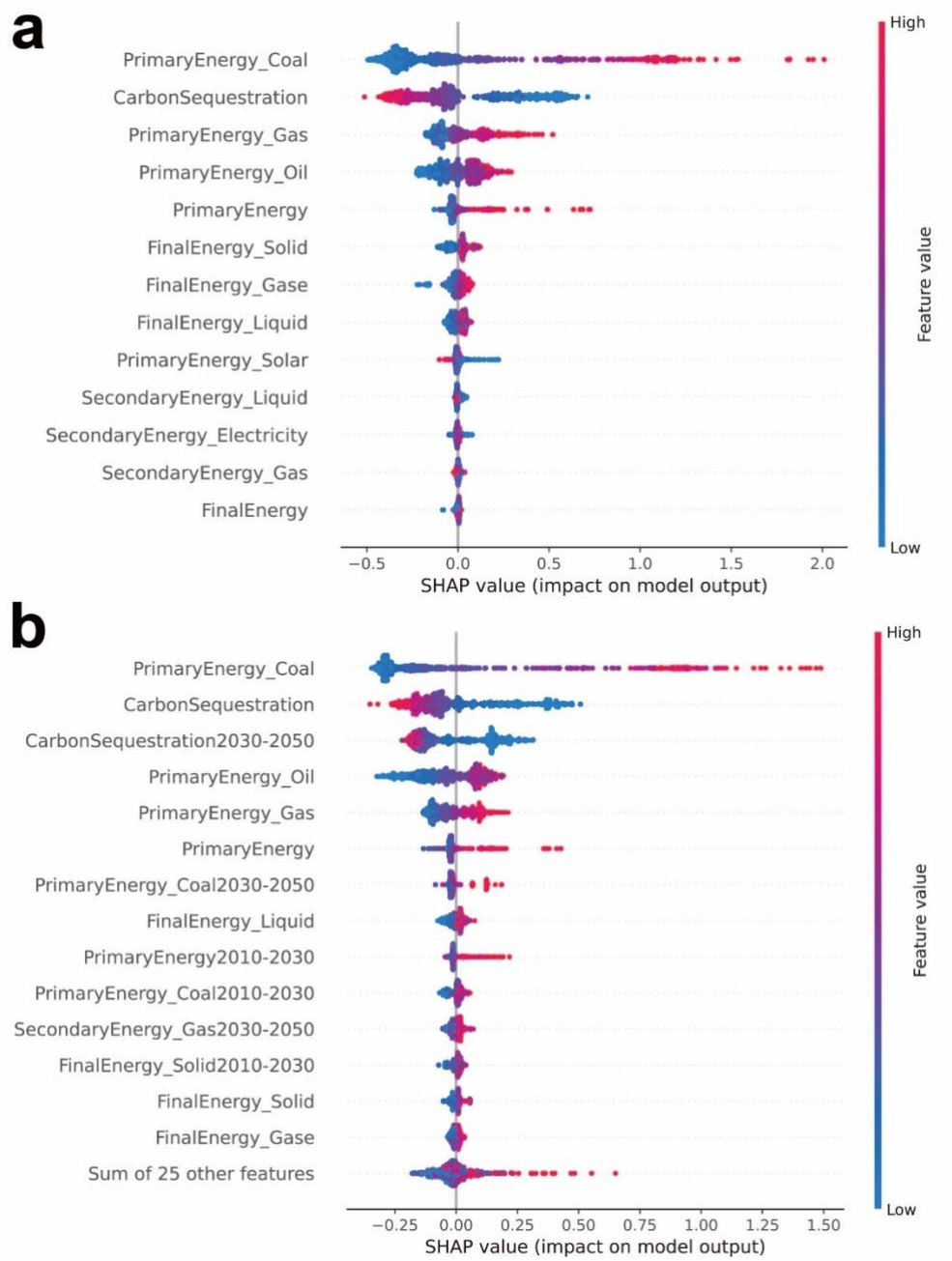


Figure S7. SHAP values of key variables, the directions and distributions indicate each variable's influence on outcomes from the C56 group as baseline. a. original 13 key variables. **b.** 39 key variables, including average change rate during 2010-2030, average change rate during 2030-3050, cumulative amount from 2010 to 2100 and model, same as Fig.2 panel b.

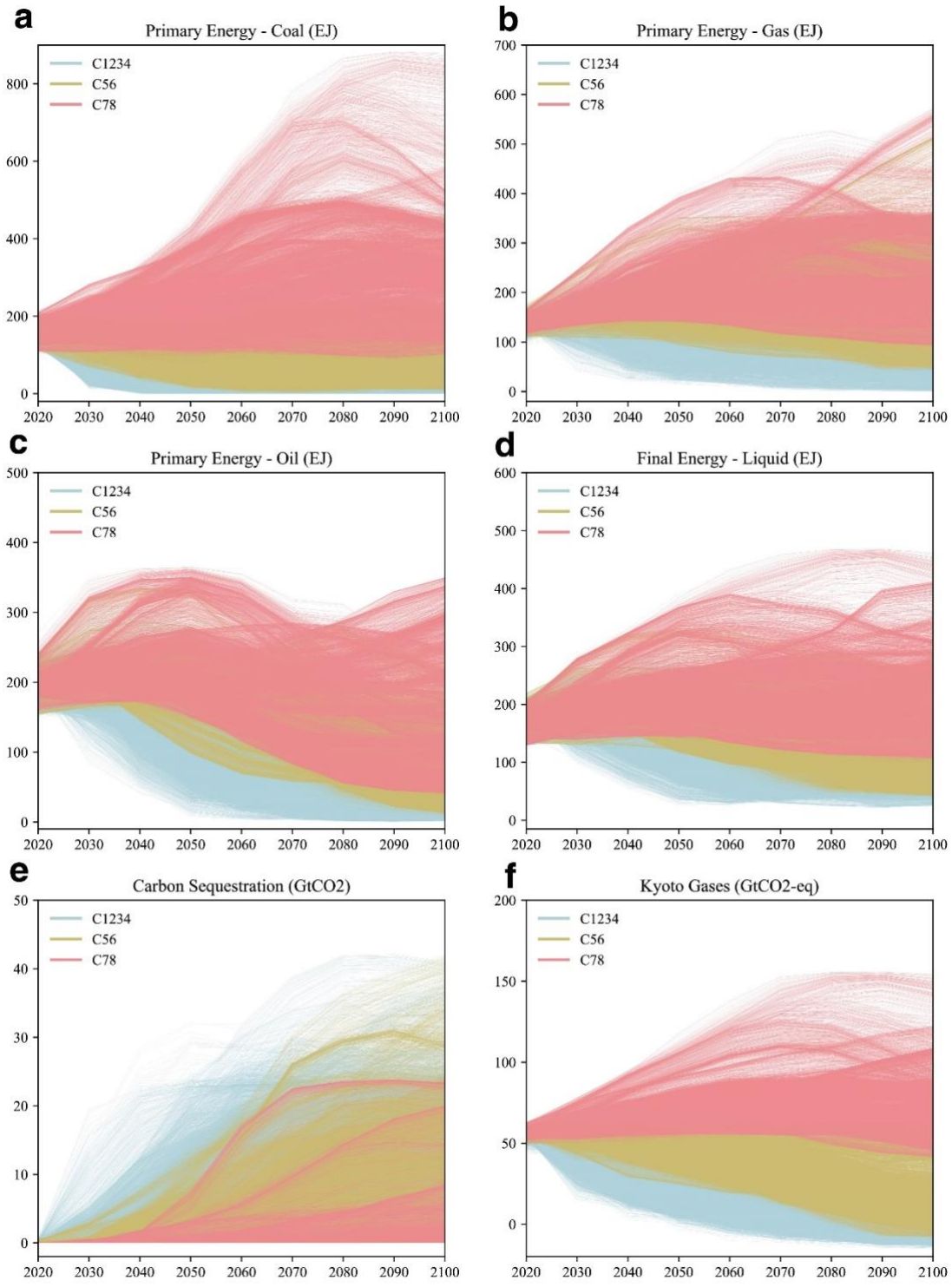


Figure S8. 30,000 IPCC-consistent synthetic scenarios generated by VAE. Synthetic time series of **a.** primary coal consumption (EJ), **b.** primary gas consumption (EJ), **c.** primary oil consumption (EJ), **d.** final liquids consumption (EJ), **e.** carbon sequestration ($MtCO_2$), and **f.** aggregated Kyoto gases (Gt CO_2-eq) by VAE.

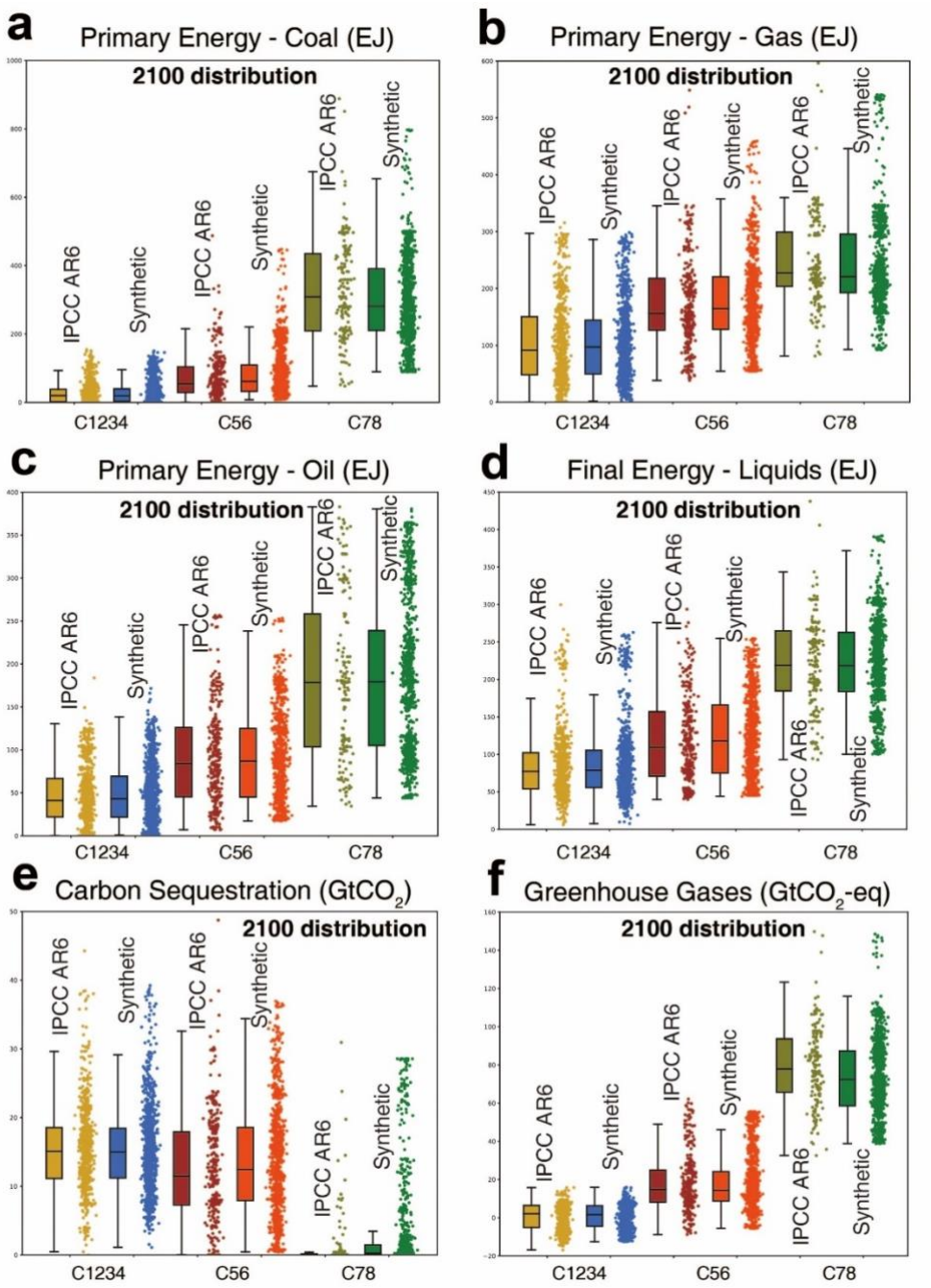


Figure S9. Distribution of key variables in IPCC AR6 and synthetic scenarios (VAE) in 2100. a. primary coal consumption, **b.** primary gas consumption, **c.** primary oil consumption, **d.** final liquids consumption, **e.** carbon sequestration, **f.** aggregated greenhouse gases The central line within each box indicates the median, while the bottom and top edges correspond to the 25th and 75th percentiles (the interquartile range, IQR). The whiskers extend to the furthest data points within 1.5 times the IQR from the lower and upper quartiles.

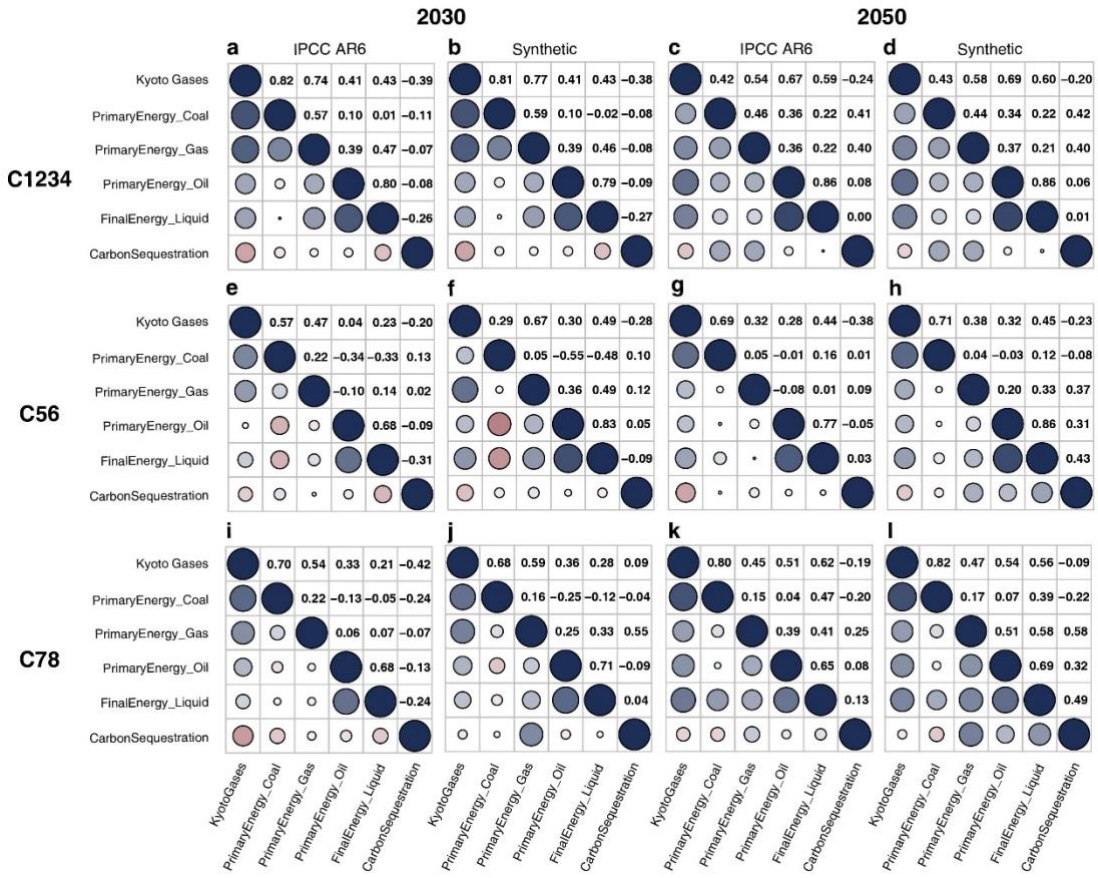


Figure S10. Corresponding heatmaps of key variables for IPCC AR6 and VAE-generated synthetic scenarios in 2030 and 2050

Supplementary Tables

Table S1. Classification of global pathways into warming levels based on IPCC AR6.

Description	Definition	Scenarios
C1: Limit warming to 1.5°C (>50%) with no or limited overshoot	Reach or exceed 1.5°C during the 21st century with a likelihood of ≤67%, and limit warming to 1.5°C in 2100 with a likelihood >50%. Limited overshoot refers to exceeding 1.5°C by up to about 0.1°C and for up to several decades.	97
C2: Return warming to 1.5°C (>50%) after a high overshoot	Exceed warming of 1.5°C during the 21st century with a likelihood of >67%, and limit warming to 1.5°C in 2100 with a likelihood of >50%. High overshoot refers to temporarily exceeding 1.5°C global warming by 0.1°C–0.3°C for up to several decades.	133
C3: Limit warming to 2°C (>67%)	Limit peak warming to 2°C throughout the 21st century with a likelihood of >67%.	311
C4: Limit warming to 2oC (>50%)	Limit peak warming to 2°C throughout the 21st century with a likelihood of >50%.	159
C5: Limit warming to 2.5°C (>50%)	Limit peak warming to 2.5°C throughout the 21st century with a likelihood of >50%.	212
C6: Limit warming to3°C (>50%)	Limit peak warming to 3°C throughout the 21st century with a likelihood of >50%.	97
C7: Limit warming to 4°C (>50%)	Limit peak warming to 4°C throughout the 21st century with a likelihood of >50%.	164
C8: Exceed warming of 4°C (≥50%)	Exceed warming of 4°C during the 21st century with a likelihood of ≥50%.	29
	Total:	1202

Table S2. Global scenarios by modelling framework and climate category. Table includes number of scenarios that passed all vetting checks. Unique model versions have been grouped into modelling frameworks for presentation in this table[1]. For a full list of unique model versions, please see the AR6 scenario database.

Model group	C1: Limit to 1.5°C (>50%) with no or limited OS	C2: Return to 1.5°C (>50%) after high OS	C3: Limit to 2°C (>67%)	C4: Limit to 2°C (>50%)	C5: Limit to 2.5°C (>50%)	C6: Limit to 3.0°C (>50%)	C7: Limit to 4.0°C (>50%)	C8: Exceed 4.0°C (≥50%)	Total with climate categorization
AIM/CGE+Hub	4	3	17	8	13	4	6	-	55
C-ROADS	3	2	-	-	-	-	-	1	6
COFFEE	1	4	14	15	21	9	1	-	65
DNE21+	-	-	-	-	-	-	-	-	-
EPPA	-	-	1	3	-	1	2	-	7
En-ROADS	-	-	-	-	-	-	-	-	-
GCAM	6	6	13	9	6	-	4	1	45
GCAM-PR	-	-	-	-	-	1	2	-	3
GEM-E3	2	10	12	6	5	3	3	-	41
GRAPE-15	-	-	-	-	-	-	-	-	-
IMAGE	7	9	34	18	22	16	34	2	142

[1] Scenario numbers by modelling framework combine submissions from different model versions of the same model (indicated by version number or project name in the AR6 scenario database). For the AIM, MESSAGE and REMIND modelling frameworks, the grouping covers the following distinct models (including different versions):
AIM/CGE+Hub: AIM/CGE, AIM/Hub
MESSAGE: MESSAGE, MESSAGE-Transport
MESSAGE-GLOBIOM: MESSAGE-GLOBIOM, MESSAGEix-GLOBIOM.
REMIND: REMIND, REMIND-H13, REMIND-Buildings, REMIND-Transport, REMIND_EU

Model group	C1: Limit to 1.5°C (>50%) with no or limited OS	C2: Return to 1.5°C (>50%) after high OS	C3: Limit to 2°C (>67%)	C4: Limit to 2°C (>50%)	C5: Limit to 2.5°C (>50%)	C6: Limit to 3.0°C (>50%)	C7: Limit to 4.0°C (>50%)	C8: Exceed 4.0°C (≥50%)	Total with climate categorization
MERGE-ETL	-	-	-	1	-	-	-	-	1
MESSAGE	-	-	-	-	-	-	-	-	-
MESSAGE-GLOBIOM	20	43	59	39	57	20	28	-	266
POLES	4	10	26	24	20	11	19	-	114
REMIND	13	12	34	1	7	6	22	9	104
REMIND-MAgPIE	28	32	50	15	27	13	26	2	193
TIAM-ECN	-	-	20	6	10	4	5	-	45
TIAM-UCL	-	-	-	-	-	-	-	-	-
TIAM-WORLD	-	-	-	-	-	-	-	-	-
WITCH	5	1	29	14	24	9	4	4	90
WITCH-GLOBIOM	4	1	2	-	-	-	8	10	25
Total	97	133	311	159	212	97	164	29	1202

Table S3. Detailed model feature variable in the model

Model	Versions	Institute
IMAGE	IMAGE 3.0 IMAGE 3.0.1 IMAGE 3.0.2 IMAGE 3.2	Netherlands Environmental Assessment Agency (PBL)
TIAM-ECN	TIAM-ECN 1.1	Netherlands Organization for applied scientific research (TNO)
GCAM	GCAM 4.2 GCAM 5.2 GCAM 5.3 GCAM-PR 5.3	Joint Global Change Research Institute (JGCRI)
POLES	POLES ADVANCE POLES CD_LINKS POLES EMF30 POLES EMF33 POLES ENGAGE POLES GECO2019	Joint Research Council (JRC)
AIM	AIM/CGE 2.0 AIM/CGE 2.1 AIM/CGE 2.2 AIM/Hub-Global 2.0	National Institute for Environmental Studies (NIES)
MESSAGE (ix-GLOBIOM)	MESSAGE-GLOBIOM 1.0 MESSAGEix-GLOBIOM 1.0 MESSAGEix-GLOBIOM 1.1 MESSAGEix-GLOBIOM 1.2 MESSAGEix-GLOBIOM_GEI 1.0	Internatinal Institue of Applied Systems Analysis (IIASA)
REMIND	REMIND 1.6 REMIND 1.7 REMIND 2.1 REMIND-Building 2.0 REMIND-MAgPIE 1.5 REMIND-MAgPIE 1.7-3.0 REMIND-MAgPIE 2.0-4.1 REMIND-MAgPIE 2.1-4.2 REMIND-MAgPIE 2.1-4.3 REMIND-Transport 2.1	Potsdam Institute for Climate Impact Research (PIK)

Model	Versions	Institute
WITCH	WITCH 4.6	Euro-Mediterranean Center for Climate Change (CMCC)
	WITCH 5.0	
	WITCH-GLOBIOM 3.1	
	WITCH-GLOBIOM 4.2	
	WITCH-GLOBIOM 4.4	
COFFEE	COFFEE 1.1	Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering (COPPE/UFRJ)
C-ROADS	C-ROADS-5.005	Climate Interactive
EPPA	EPPA 6	Massachusetts Institute of Technology (MIT)
GEM-E3	GEM-E3_V2021	Institute of Communication and Computer Systems (ICCS)
MERGE-ETL	MERGE-ETL 6.0	Paul Scherrer Institute (PSI)

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